

Windows 10 Pro – Security Hardening

Project Overview:

This project demonstrates how to harden a **Windows 10** Pro system against common security threats using **BitLocker**, **Windows Defender**, and **Firewall rules**. System hardening is an essential IT Support skill, ensuring that endpoints are secured, compliant, and resilient against attacks.

Objectives

- Enable and configure **BitLocker drive encryption**
- Manage **Windows Defender Antivirus settings**
- Apply **Firewall rules** to control applications and services
- Test and validate the security settings

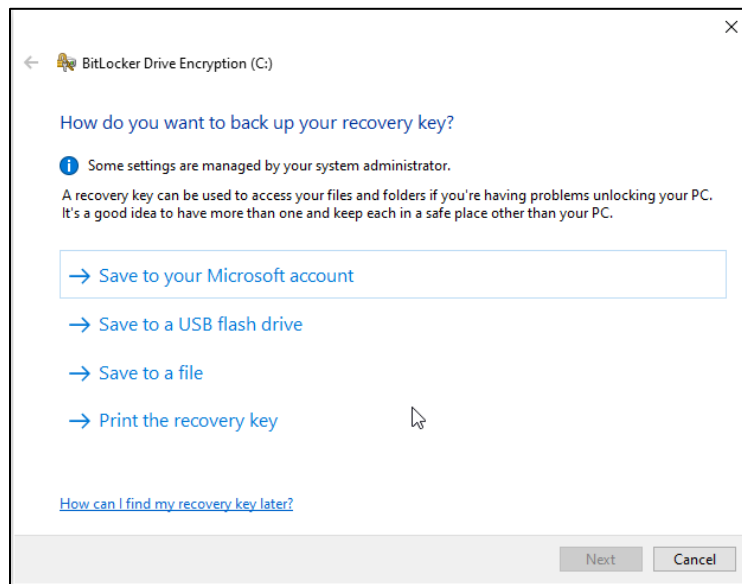
Tools & Skills Used

1. **Windows 10 Pro** Virtual Machine.
2. **BitLocker Drive Encryption**
3. **Windows Defender Antivirus**
4. **Windows Defender Firewall**
5. Skills: Endpoint Security · Troubleshooting · Documentation

Steps & Implementation

1. Enable BitLocker Drive Encryption

- Opened **Control Panel** → **BitLocker Drive Encryption**
- Enabled BitLocker on the `C:` system drive
- Configured with a **password unlock** method and saved recovery key



2. Verify BitLocker Status

- Ran command in powershell: `manage-bde -status`
- Verified encryption progress and protection status

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Administrator: Windows PowerShell

Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> manage-bde -status
BitLocker Drive Encryption: Configuration Tool version 10.0.19041
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Disk volumes that can be protected with
BitLocker Drive Encryption:
Volume E: [Backup]
[Data Volume]

    Size:                50.00 GB
    BitLocker Version:    None
    Conversion Status:    Fully Decrypted
    Percentage Encrypted: 0.0%
    Encryption Method:    None
    Protection Status:    Protection Off
    Lock Status:          Unlocked
    Identification Field: None
    Automatic Unlock:     Disabled
    Key Protectors:       None Found

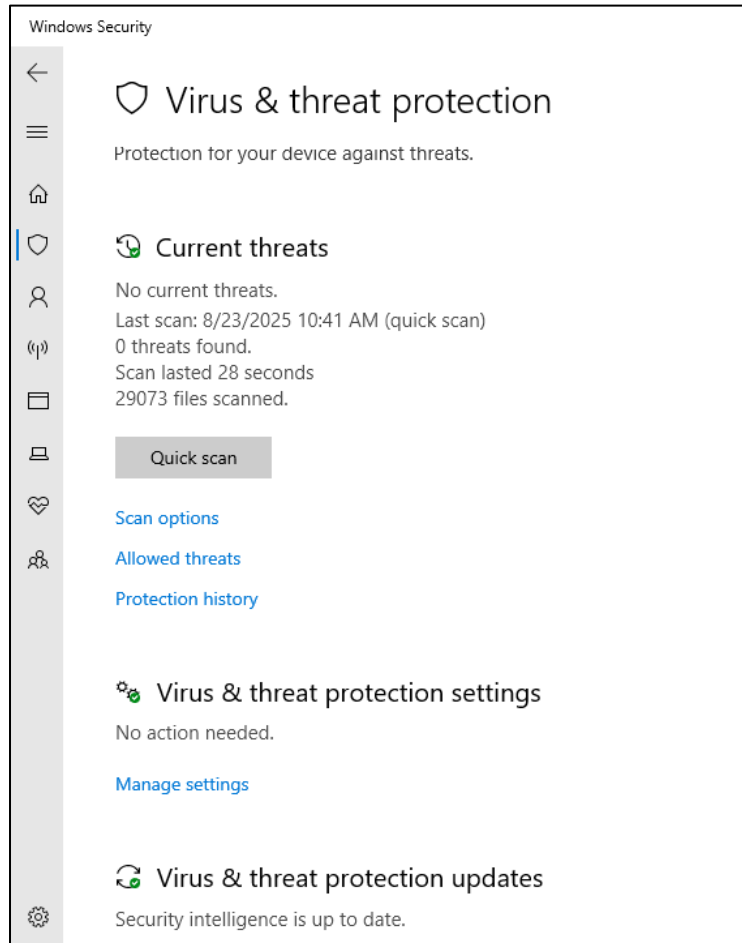
Volume C: []
[OS Volume]

    Size:                49.44 GB
    BitLocker Version:    2.0
    Conversion Status:    Used Space Only Encrypted
    Percentage Encrypted: 100.0%
    Encryption Method:    XTS-AES 128
    Protection Status:    Protection On
    Lock Status:          Unlocked
    Identification Field: Unknown
    Key Protectors:
        Password
        Numerical Password

PS C:\Windows\system32>
```

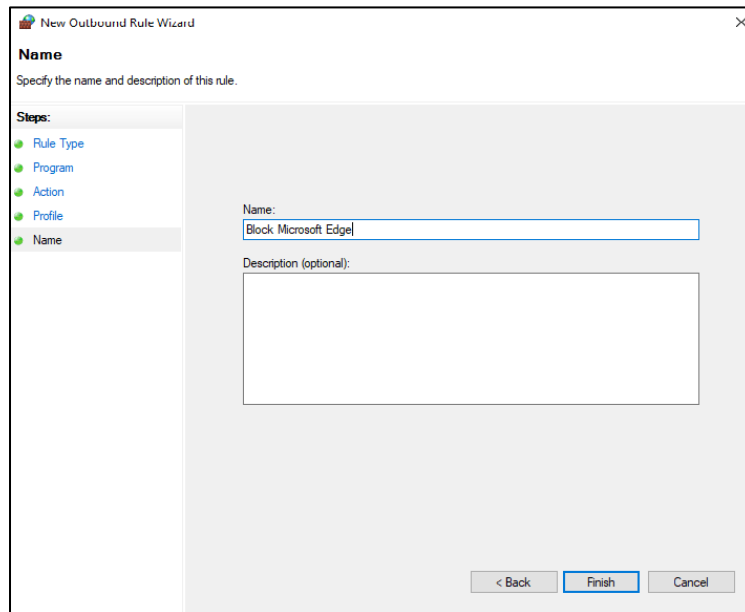
3. Configure Window Defender Antivirus

- Opened **Windows Security** → **Virus & Threat Protection**
- Turned on **Real-time protection** and **Cloud-delivered protection**
- Updated definitions manually



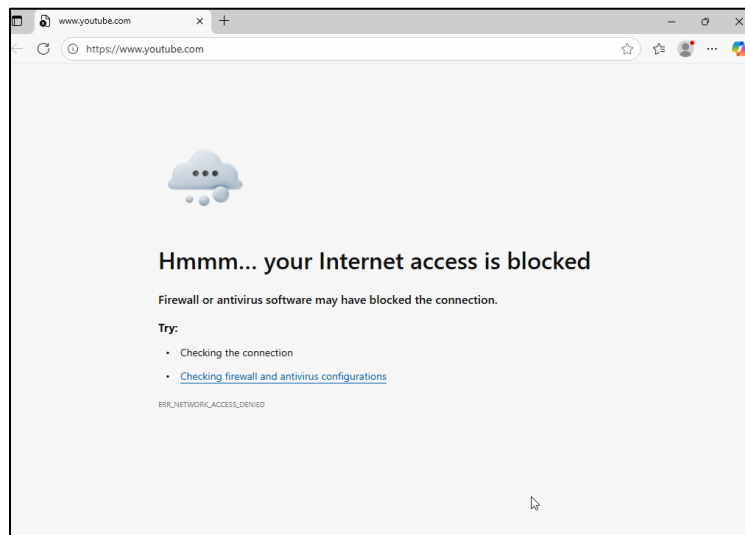
4. Create Firewall Rules

- Opened **Windows Defender Firewall** → **Advanced Settings**
- Created an **Outbound Rule** to block a Microsoft Edge



5. Test Firewall Rules

- Attempted to run blocked application → Access denied by firewall



Result:

- ✓ Successfully enabled and verified **BitLocker encryption**
- ✓ Configured **Windows Defender Antivirus** for real-time protection
- ✓ Applied and tested **an outbound firewall rule**
- ✓ Improved endpoint security by enforcing encryption, antivirus, and network rules

What I Learned:

- Learned how to secure Windows 10 endpoints with BitLocker and Defender
- Practiced applying and testing firewall rules
- Demonstrated IT Support readiness in **endpoint hardening & security troubleshooting**